among any number between 10 and 20. When we say "variability", we implicitly talk about our uncertainty regarding how the test accuracy of our model behaves as a function of the number of layers. Even though we know 10 layers lead to 81% and 20 layers lead to 75%, we cannot say for certain what value, say, 15 layers would yield. This is to say we need to account for our level of uncertainty when considering these values between 10 and 20.

Further, what if some number greater than 20 will actually give us the highest accuracy possible? This is the case for many large data sets, where a sufficient depth is necessary for a neural net to learn anything useful. Or, though unlikely, what if a small number of layers (fewer than 5) is actually what we need?

How should we explore these different options in a principled way, so that when our time runs out and we have to report back to our boss, we can be sufficiently confident that we have arrived at the best number of layers for our model? This question is an example of the general problem called "expensive black-box optimization," which we discuss next.

1.1.2 The problem of expensive, black-box optimization

We will now formally introduce the problem of expensive, black-box optimization, which is what Bayesian optimization aims to solve. Understanding why this is such a difficult problem will help us understand why Bayesian optimization is preferred over simpler, more naïve approaches.

In this problem, we have black-box access to a function (some input–output mechanism), and our task is to find the input that maximizes the output of this function. The function is often called the *objective function*, as optimizing it is our objective, and what we want to find is the *optimum* of the objective function, the input that yields the highest function value.

> **NOTE** **Characteristics of the objective function**
>
> **The term "black-box" means that we don't know what the underlying formula of the objective is; all we have access to is the function output when we make an observation by computing the function value at some input. In our neural net example, we don't know how the accuracy of our model will change if we increase the number of layers one by one (otherwise we would just pick the best one).**
>
> **The problem is "expensive" because in many cases, making an observation (evaluating the objective at some location) is prohibitively costly, rendering a naïve approach such as an exhaustive search intractable. In ML and especially deep learning, time is usually the main constraint, as we already discussed.**

Hyperparameter tuning belongs to this class of expensive black-box optimization problems, but it is not the only one! Imagine any procedure in which we are trying to find some

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